**SET - A**

```
class ExpressDelivery  extends Delivery{
  public String type;
  public double cost;
  public int days;
  public ExpressDelivery(String type, String packageId, double weight, double distance){
    super(packageId, weight, distance);
    this.type = type;
  }
   public void calculateCost(){
     if(type.equals("Fast")){
       cost = super.baseFare()+ weight * 7 + distance * 3;
     }
     else{
       cost = super.baseFare()+ weight * 5 + distance * 1;
     }
  }
  public void estimatedDays(){
    if(distance < 50){
      days = 2;
    }
    else if (distance < 70){
      days = 3;
    }
    else{
      days = 4;
    }
  }
  public void packageInfo() {
        printDetails();
        System.out.println("Type: " + type);
        System.out.println("Cost: " + cost + "Tk");
        System.out.println("Estimated Days: " + days);
    }
}
```

**SET - B**

```
class PremiumTicket extends Ticket {
    String type;
    double cost;
    String category;

    public PremiumTicket(String type, String ticketId, int seats, double duration) {
        super(ticketId, seats, duration);
        this.type = type;
    }

    public void calculateCost() {
        if(type.equals("VIP")) {
            cost = super.baseFare() + (seats * 500) + (duration * 100);
        }
        else {
            cost = super.baseFare() + (seats * 300) + (duration * 50);
        }
    }

    public void estimatedCategory() {
        if(duration < 3) {
            category = "Short Event";
        }
        else if(duration <= 5) {
            category = "Medium Event";
        }
        else {
            category = "Long Event";
        }
    }

    public void ticketInfo() {
        printDetails();
        System.out.println("Type: " + type);
        System.out.println("Cost: " + cost + "Tk");
        System.out.println("Category: " + category);
    }
}
```

# Marking Rubric

**For both sets:**

| Criteria | Points |
|---|---|
| Basic Class Design<br>→ **[0.5]** Correct class declaration with extends Ticket<br>→ **[0.5]** Proper instance variables (type, cost, category) | 1 |
| Constructor: **ExpressDelivery() / PremiumTicket(String)**<br>→ **[0.25]** Correct constructor syntax<br>→ **[0.5]** Correct use of super() to initialize parent variables<br>→ **[0.25]** Assign type properly | 1.5 |
| Method: **calculateCost(): void**<br>→ **[0.5]** Correct method declaration syntax<br>→ **[1]** Correct conditional check for type (Fast/ Saver) or (VIP / Regular)<br>→ **[1]** Correct cost calculation using formula | 2.5 |
| Method: **estimatedDays()**:void / **estimatedCategory()**:void<br>→ **[0.5]** Correct method declaration syntax<br>→ **[1.5]** Correct conditional logic for distance/duration ranges<br>→ **[0.5]** Assign correct days/category | 2.5 |
| Method: **packageInfo()**: void / **ticketInfo():** void<br>→ **[0.5]** Correct method declaration syntax<br>→ **[0.5]** Call to printDetails()<br>→ **[1]** Correctly print type, cost, and days / category<br>→ **[0.5]** Proper formatting of output | 2.5 |
| **Total:** | **10** |

Some students may have printed all the infos inside calculateCost() method. If the output matches then you may deduct 0.25 marks for not printing in info() method.